

Computer Networks: Introduction

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This Lecture

- ☐ Administrative stuff
- ☐ What is Network ?
- ☐ What Network Does ?
- ☐ Why Study Networking ?
- ☐ How Network Looks Like ?
- ☐ Historical Developments
- ☐ How to Study Networking ?



Administrative Stuff

☐ Textbooks

1. Jim Kurose and Keith W Ross, “Computer Networking: A Top-down Approach”, 8th edition 2022
2. P. Kathiravelu and M. O. Faruque Sarker, “Python Programming Cookbook”, Packt Publishing, 2nd edition, 2023

☐ Reference Books

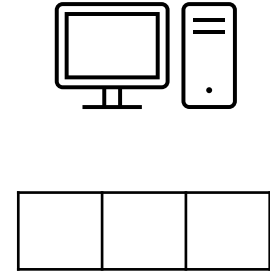
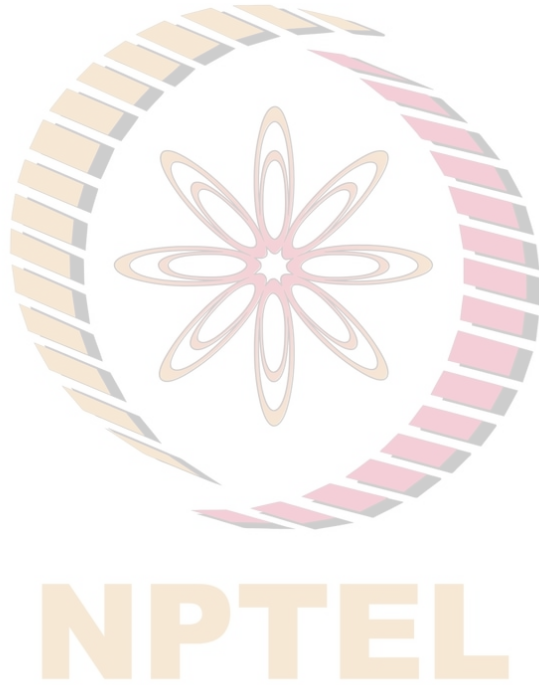
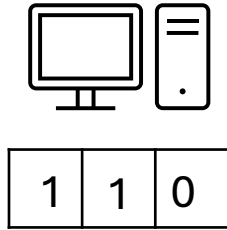
1. Larry Peterson and Bruce Davie, "Computer Networks, A Systems Approach", Morgan Kaufmann Publishers In, 6th edition 2021
2. W. Richard Stevens, "Unix Network Programming: The Sockets Networking API", Pearson Education, 3rd ed. 2017

1. What is Network ?

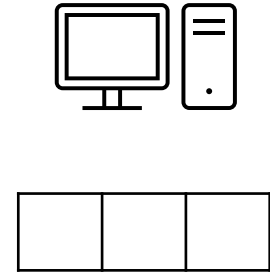
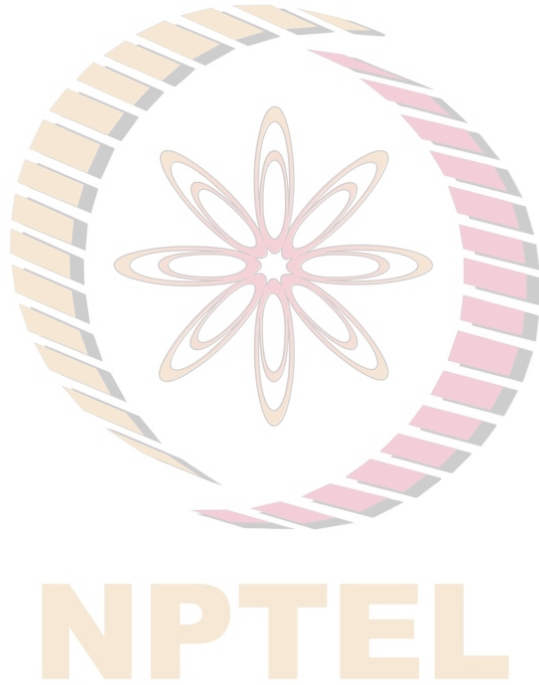
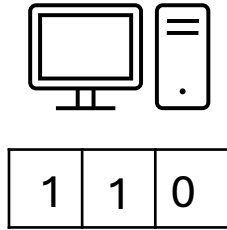
- ❑ *An interconnection of computer systems to transfer data between the applications running in these systems*



2. What the Network Does ?



2. What Network Does ?



3. Why Study Networking ?

- ☐ Many reasons to study
 - ☐ Web browsing
 - ☐ Email
 - ☐ Video conferencing
 - ☐ Audio call (VoIP)
 - ☐ Remote desktop connection
 - ☐ Many more



3. Why Study Networking ?



Amazon Echo



Internet refrigerator



Security Camera



Internet phones



Gaming devices



IP picture frame



Slingbox: remote control cable TV



Pacemaker & Monitor



Web-enabled toaster + weather forecaster



sensorized, bed mattress



AR devices



Fitbit



Tweet-a-watt:
monitor energy use



bikes



cars

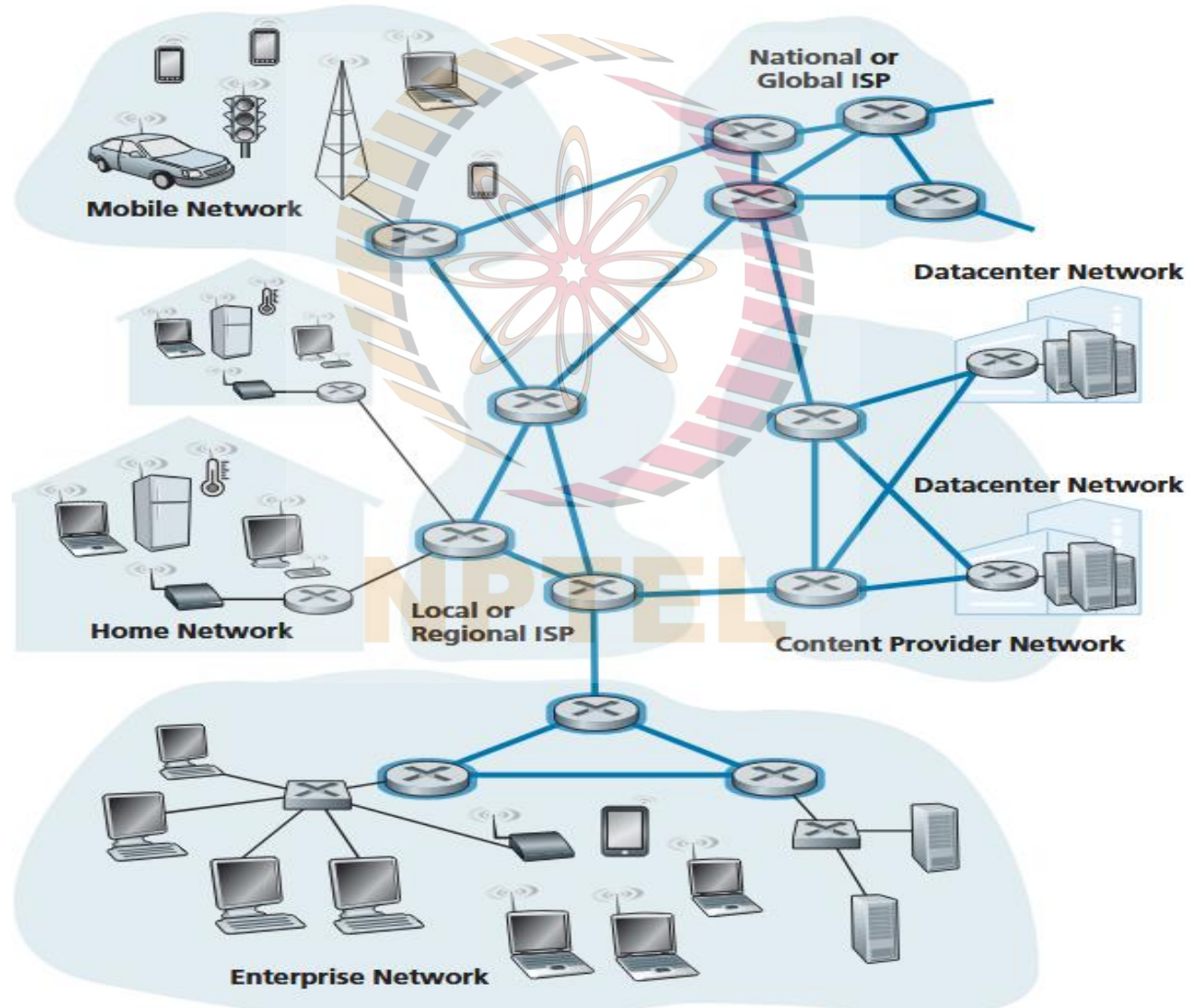


scooters

4. How Computer Networks Look Like ?

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4. How Networks Look Like ?



5. History of Networking

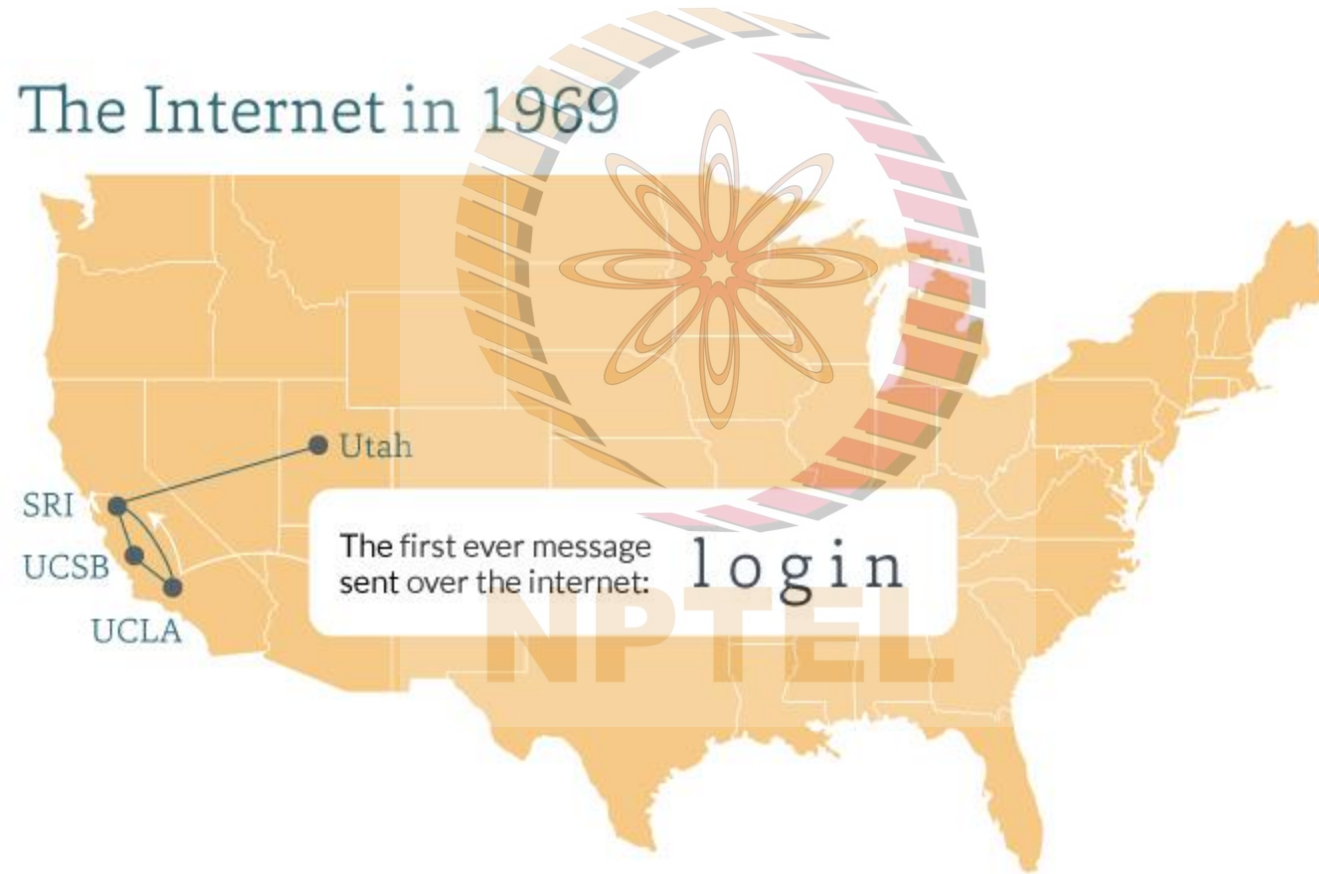
- ❑ 1959 – ARPA
- ❑ 1960- 1966 Theoretical research
- ❑ 1966 – ARPANET formally proposed a practical network
- ❑ 1968 - Bolt Beranek and Newman (BBN) was tasked to design interfacing processors
- ❑ **October 29, 1969: First ARPANET Connection UCLA ↔ SRI**

5. History of Networking

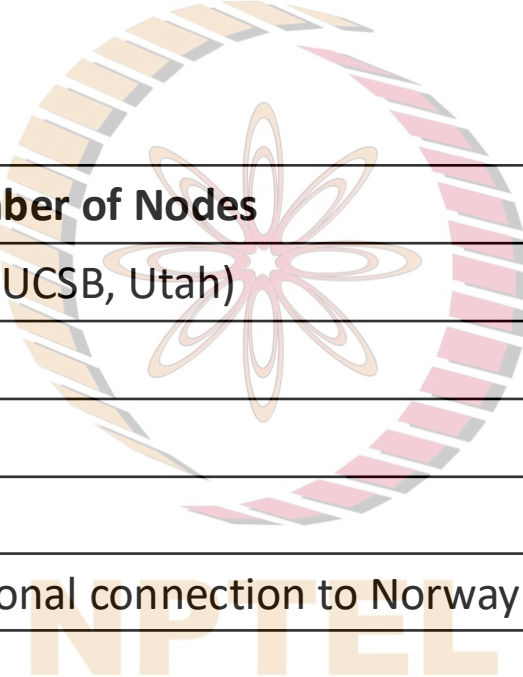


Date	Institution	Location	Notes
October 29, 1969	UCLA	Los Angeles, CA	First ARPANET node; first message sent to SRI
Early November 1969	SRI (Stanford Research Institute)	Menlo Park, CA	Second node
Late November 1969	UC Santa Barbara (UCSB)	Santa Barbara, CA	Third node
December 1969	University of Utah	Salt Lake City, UT	Fourth node

5. History of Networking



5. History of Networking: Growth of ARPANET



Year	Approx. Number of Nodes
1969	4 (UCLA, SRI, UCSB, Utah)
1970	13
1971	18
1972	23
1973	first international connection to Norway and the UK

5. History of Networking: Private Networks

- ❑ 1974 - **SNA** Systems Network Architecture by IBM
- ❑ 1975 - **DECnet** by Digital Equipment Corporation
- ❑ 1975 - **Telenet** by Bolt Beranek and Newman (BBN)
- ❑ 1977 - **XNS** Xerox Network Systems
- ❑ 1981 - **BITNET** (Because It's Time Network)

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